OBI-WAN: An AI Advising System for Human–AI Collaborative Education-to-Career Decisions

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Abstract. College students and early-career individuals face complex academic and workforce decisions amid rapid change driven by large language models (LLMs). Although AI-enabled advising systems can support equitable, informed choices at scale, many lack interpretability and meaningful human–AI collaboration. This paper introduces the prototype implementation of the Occupation Based Index for Workforce AI Indicator (OBI-WAN), a system-oriented AI advising framework that integrates intermediate work activities, work context constraints data from Occupational Information Network and program data from Integrated Postsecondary Education Data System to support pathway exploration across education and work. OBI-WAN operationalizes a context-adjusted AI applicability score that combines task centrality, LLM exposure classification, and contextual feasibility of AI deployment within a specified scoring model. This framework was implemented within an agentic, multi-turn Retrieval-Augmented Generation (RAG) architecture designed to elicit learner preferences and generate transparent, explanation-driven guidance. Initial validation via system-level outputs and design-based interaction scenarios demonstrates how OBI-WAN differentiates occupations into low, hybrid, and high AI integration zones and supports interpretable reasoning across occupations, skills, and educational programs without presenting AI impact as a binary or deterministic outcome. This work contributes an architecture-driven proof-of-concept that positions AI as a collaborative teammate. Implications are discussed for explainable AI, human–AI collaborative advising, and the design of scalable, interpretable AI-enabled infrastructures that preserve learner agency amid AI-driven workforce change.

Keywords: LLMs, Retrieval-Augmented Generation, Education-to-Career Pathways, Human–AI Collaboration, eXplainable AI, Proof-of-Concept.

1 Introduction

Learners today face increasingly complex decisions as they move from education into the labor market. Post-secondary advising offices often lack sufficient resources to provide individualized, data-driven guidance at scale, contributing to inequities in access to actionable information. Simultaneously, the rapid integration of large language models (LLMs) into workplace tasks has introduced uncertainty about future skill demands and long-term career viability. Many students struggle to interpret occupational projections, understand how AI reshapes work, or align postsecondary programs with evolving workforce needs.

Research in learning sciences and human–AI collaboration suggests that effective decision support systems should scaffold metacognitive reasoning and function as collaborative partners rather than static tools. Educational data mining and pathway

modeling further emphasize the value of integrating academic, skills, and workforce data to support holistic guidance.

This paper introduces the Occupation Based Index for Workforce AI Indicator (OBI-WAN), an AI-augmented advising system that models pathway decision-making through transparent human–AI collaboration. The system integrates O*NET task data, work context variables, and IPEDS program conferral patterns to compute an interpretable AI applicability score estimating how AI may reshape occupational tasks. OBI-WAN uses a multi-turn conversational flow to gather learner preferences, retrieve aligned programs, and generate interpretable explanations of occupational exposure and program alignment. The occupation-based index serves as the semantic grounding layer within a RAG system, contextualizing AI applicability within broader occupational competencies and institutional program data to support holistic, pathway-level reasoning.

This work contributes:

1) A context-adjusted AI applicability score combining task centrality, LLM exposure, and work context-based operational feasibility
2) A unified pipeline linking educational programs with occupational data and AI applicability scores
3) An AI advising system that functions as a human–AI teamwork, supporting collaborative decision-making.

2 Related Research

2.1 Task-Based and Context-Based Modeling of AI Exposure

Task-based frameworks have been widely used to quantify AI automation risk by examining which job tasks are susceptible to AI augmentation or substitution [1, 2]. However, the actual impact of AI deployment is constrained by work context, including physical interaction requirements, interpersonal demands, and organizational structures [2]. OBI-WAN integrates both task-level and context-level modeling to produce a more realistic measure of occupational AI applicability.

2.2 Recommender Systems for Education-to-Career Pathways

Educational recommender systems traditionally support course selection, instructional sequencing, or personalized learning environments [3]. Yet, few systems integrate federal labor datasets with post-secondary program data or provide transparent explanations of how AI-related changes may influence future occupational trajectories. OBI-WAN fills this gap by merging structured labor data with educational data mining techniques.

2.3 Human–AI Collaboration and Explainability

Explainable AI (XAI) emphasizes transparency in how AI systems derive their outputs, enabling users to understand, contest, or meaningfully rely on system recommendations [4]. In educational contexts, explainability is essential for supporting equitable advising

practices and ensuring that learners remain central decision-makers [4]. OBI-WAN adopts an interpretable scoring model. The system enables the agent to contextualize the AI applicability score by retrieving granular competency profiles (skills, knowledge, abilities) of each occupation and aligned educational pathways or vice versa.

3 Methodology

3.1 System Architecture

The system is implemented as an autonomous agentic framework using the Google AI Development Kit (ADK). A Gemini 2.5 Flash LLM manages multi-turn dialogue state and intent routing to specialized tools. Semantic retrieval uses text-embedding-004 and cosine similarity to resolve natural language queries into standardized O*NET and CIP codes. Custom tools retrieve occupational competencies, compute AI applicability scores, perform CIP–O*NET relational joins, and filter institutional data based on user constraints. The agent is deployed via the ADK Web Runner as a stateful web interface.

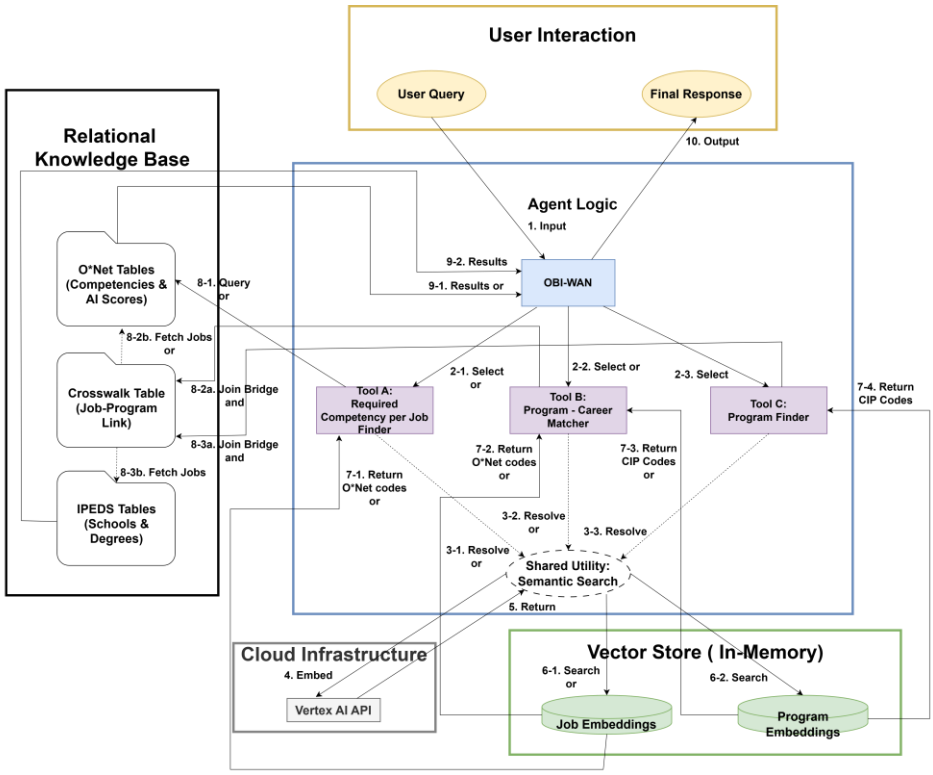

Fig 1. System Architecture Diagram

3.2 Data Structure and Knowledge Representation

The knowledge base links O*NET occupational codes to competencies and AI applicability scores, and CIP codes to institutional program data from IPEDS. Data are structured as occupation, competency, program, and institution nodes connected via relational edges within BigQuery. Precomputed embeddings enable semantic matching between user queries and standardized occupational or program identifiers.

3.3 AI Applicability Modeling

OBI-WAN integrates O*Net (Intermediate Work Activities, Work Context, skills, knowledge, abilities) and IPEDS (program list and conferral) data to model occupational AI applicability. These datasets enable mapping between competencies required for each occupation, task demands, contextual constraints, and educational pathways.

Each IWA was classified into exposure levels, E0 (no exposure), E1 (direct exposure), and E2 (LLM + tools exposure), following Eloundou et al. [2]. Numeric coding was assigned as E0 = 0, E1 = 1, and E2 = 0.5. The same rubric was applied to Work Context variables. However, because context constrains rather than enables deployment, values were recoded as E0 = 1, E1= 0, and E2 = 0.5. This inverse coding distinguishes theoretical task compatibility from operational feasibility.

Task centrality for task *t* within IWA *i* is:

$$C_{it} = 2^{I_{it}} * R_{it}$$

where I_{it} denotes the O*Net importance rating of task *t* within each occupation *o*, and R_{it} represents the corresponding relevance rating.

The weight of each IWA within occupation *o* are:

$$W_{oi} = \frac{\sum_{t \in i} C_{it}}{\sum_j \sum_{t \in j} C_{it}}$$

The occupation-level IWA exposure score is:

$$S_o = \sum_{i \in I_o} W_{oi} \cdot E_{k(i)}$$

Work context variables capture environmental and interpersonal conditions that affect LLM deployment. For occupation *o*, the raw constraint score is:

$$R_o = \sum_{c \in C_o} x_{oc} \cdot \gamma(c)$$

where x_{oc} is the context frequency value (1–5 scale), C_o is the set of work contexts associated with occupation *o*, and $\gamma(c)$ is the inversely coded exposure value.

To normalize across occupations with varying numbers of context variables, min–max scaling is applied:

$$\Gamma_o = \frac{R_o - R_{min,o}}{R_{max,o} - R_{min,o}}$$

yielding a constraint score $\Gamma_o \in [0, 1]$.

The final context-adjusted AI applicability score integrates theoretical exposure and contexture feasibility:

$$A_o = S_o \cdot (1 - \Gamma_o)$$

This multiplicative formulation ensures that occupations with high LLM-compatible task content but strong contextual constraints receive moderated scores. Kernel density

estimation of the resulting distribution revealed local minima near 0.2 and 0.5, which define three AI Applicability Zones: Human-Centric (< 0.2), Hybrid (0.2–0.5), and High Integration (> 0.5). These zones structure explanation logic within the advising agent.

3.4 Retrieval and Generation Mechanism

The system employs an agentic Retrieval-Augmented Generation (RAG) loop governed by three sequential stages.

Intent routing and state management. A Gemini 2.5 Flash engine handles intent routing by classifying queries and directing them to each tool (see Fig 1 Tool A–C). For underspecified inputs, a multi-turn dialog manager elicits missing parameters, such as modality or degree level, while conversational state is maintained via a ToolContext that persists entity identifiers across the session.

Dual-stage retrieval. The agent executes a two-part process: entity resolution matches query-based embeddings against vector registries to establish a standardized code anchor, after which graph traversal utilizes parameterized SQL queries to perform precise relational joins (e.g., CIP to O*NET) across the crosswalk bridge.

Explanation-driven generation. The LLM synthesizes this "Ground Truth" into transparent, goal-oriented guidance. Rather than delivering raw data, the agent provides an auditable narrative that explains the rationale behind specific program-to-career alignments and AI-risk profiles. This grounding mechanism constrains generative output to verifiable structured data, reducing hallucination and improving reliability of pathway recommendations.

4 Preliminary Evaluation via Synthetic Personas

To illustrate how OBI-WAN supports pathway exploration, synthetic learner personas with specific educational backgrounds and goals [5]. These personas were used to simulate representative interaction scenarios. Each case demonstrates the agent's ability to (a) classify occupations into AI Applicability Zones, (b) retrieve relevant career or program information, and (c) generate explanations grounded in the underlying scoring model.

4.1 Persona 1: Learner Interested in Majoring in Nursing and Exploring Career Options

In response to the query, "I'm majoring in Nursing, other than a registered nurse, what career pathways can I have?", OBI-WAN retrieves the top program matches (e.g., Nursing Practice, Registered Nursing, Nursing Education) based on semantic similarity and retrieves the occupations linked to those programs. The system then stratifies career options using the AI Applicability Score into three zones: High Integration (> 0.5), Hybrid (0.2–0.5), and Human-Centric (< 0.2).

Administrative and instructional roles (e.g., nursing instructors, health services managers) fall within the High Integration Zone, reflecting substantial AI-augmentable

documentation and managerial tasks. Clinical and embodied roles (e.g., registered nurses, acute care nurses, critical care nurses) cluster within the Hybrid Zone, where interpersonal and safety-critical demands constrain automation. No roles fall within the Human-Centric threshold in this example. This case illustrates how the score differentiates occupational trajectories within a single academic pathway.

4.2 Persona 2: Learner Seeking Skills and Program Pathways for Becoming a Mental Health Counselor

In a second scenario, a learner asks, "What skills do I need for a mental health counselor?" OBI-WAN provides the top 15 skills ranked by their O*NET importance ratings (1–5 scale), including social perceptiveness, active listening, judgment and decision-making, and persuasion. When the learner follows up with, "Which institutions offer programs to strengthen these skills?", the agent refines retrieval by eliciting degree level and modality preferences. After specification (e.g., Master's, Online), OBI-WAN returns the top institutions, including state information, ranked by conferral volumes. This scenario demonstrates multi-turn dialogue, preference-conditioned retrieval, and transparent skill- and program-level guidance.

5 Discussion and Conclusion

OBI-WAN demonstrates how an AI-augmented advising system can support human–AI collaborative reasoning during pathway exploration. By decomposing exposure into interpretable components, the system enhances user agency and reduces black-box opacity common in LLM-driven tools. For institutions, OBI-WAN provides an extensible infrastructure for institutional analytics, enabling alignment between academic offerings and emerging workforce trends. The system further contributes to accuracy in guidance by grounding recommendations in transparent federal datasets.

To build upon this proof-of-concept, synthetic personas will be refined to better reflect specific target audiences, establishing this method as a robust evaluation layer for rigorously assessing model performance. Following this system-level validation, future work includes conducting larger empirical user studies analyzing decision-making behaviors in human–AI collaboration, cross-cultural evaluations of advising narratives, examinations of users' perceptions of the agent's effectiveness in improving decision-making, and investigations into user trust, transparency, and the ways in which outcomes are moderated by digital literacy.

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